Autotrader Car Price Prediction Using Machine Learning Concept

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Abstract. This paper explores the application of advanced machine learning algorithms for predicting car prices listed on Autotrader. The primary objective is to develop a robust predictive model capable of accurately estimating the market value of a vehicle based on a comprehensive set of features, including make, model, year, mileage, condition, multimedia infotainment, multi-sensor wireless connectivity capabilities, and other disruptive functionalities. The methodology encompasses systematic data collection, meticulous preprocessing, strategic model selection, rigorous training, and thorough evaluation of various machine learning techniques. The findings of this research demonstrate the efficacy of these algorithms in enhancing price estimation accuracy within the automotive market. Quantitative performance metrics indicate that the proposed model achieves an accuracy improvement of 15% compared to traditional prediction methods. These insights are valuable for both consumers and industry stakeholders, reflecting the expanding modern and emerging trade space of car utility parameters.

Keywords: Autotrader Car Price · Machine Learning· Design efficiency.

1 Introduction

The automotive industry has witnessed a significant transformation over the past few decades, driven by advancements in technology, data analytics, and machine learning [1], [2]. One of the critical aspects of the automotive market is the pricing of used cars, which has traditionally been a complex and subjective process [3]. Accurate pricing is essential for both buyers and sellers to make informed decisions and ensure fair transactions. With the rise of online platforms such as Autotrader, which aggregate vast amounts of data on used cars, there is a growing opportunity to leverage machine learning techniques to predict car prices more accurately [1].

The traditional methods of car price estimation often rely on manual assessments and basic statistical models, which can be limited in their ability to handle the myriad factors influencing car prices [1], [3], [4]. These factors include make and model, year of manufacture, mileage, condition, location, and various other attributes that collectively determine a car's market value. The subjective nature of these assessments can lead to inconsistencies and inaccuracies, affecting market dynamics and consumer trust.

Machine learning (ML), a subset of artificial intelligence (AI), offers powerful tools and algorithms that can analyze large datasets, identify patterns, and make predictions with high accuracy [5], [6]. By applying machine learning to car price prediction, we can automate the estimation process, reduce biases, and improve the reliability of price evaluations [3]. This approach not only benefits consumers by providing fair and transparent pricing but also aids dealers and platforms like Autotrader in enhancing their service offerings.

The automotive market is dynamic and influenced by numerous factors, making price prediction a complex task. Accurate price prediction models can benefit buyers, sellers, and dealers by providing a reliable estimate of a car's market value. This study leverages machine learning techniques to develop a predictive model using data from Autotrader, one of the largest online car marketplaces.

2 Literature Review

The application of machine learning (ML) techniques in the automotive industry has grown significantly, particularly in the domain of car price prediction [3], [1]. With the advent of online car trading platforms like Autotrader, which aggregate extensive datasets of car listings, there is a unique opportunity to enhance the accuracy and reliability of car price estimations using advanced ML algorithms [7], [1]. Numerous studies have demonstrated the application of machine learning in various fields, including real estate, stock market, and automotive industries. Previous research in car price prediction has utilized regression models, decision trees, and ensemble methods [8], [9]. This study builds upon these foundations by comparing multiple machine learning algorithms and selecting the most effective model for car price prediction.

The trade space for the prediction of future car prices by buyers will undoubtedly be driven by advanced technological innovations and holistic functionalities to sustain the future lifestyles. The 6G networks are complex systems spanning many domains and enabling AI integration to boost nodes and network performance. Terrestrial and non-terrestrial networks will add edge intelligence and serve many new user equipment types (including vehicles and Internet of Things devices). For instance, the integrated next-generation satellite-cellular 5G/6G network will depend on a conglomerate of raw compute performance and massive information available among individual vehicles and infrastructure to achieve a true capability-based autonomous operation of vehicles [10]. The Third Generation Partnership Project (3GPP) has set communication standards to harmonise vehicle-to-vehicle (V2V); Vehicle-to-infrastructure (V2I); and vehicle-to-anything (V2X) connectivities [11], [12], [13], [14], [15]. The terrestrial and non-terrestrial convergence network ecosystem will utilise cutting-edge technologies including mmWave 5G New Radio (NR) / 6G mid-band to provide gigabit speeds at ultra-low latency for automotives with point-of-care and telemedicine and other seamless ubiquitous capabilities [16], [17], [18], [19]. Future car buyers will heavily rely on AI/ML algorithms to predict car prices for vehicles with large connectivity datasets including (i) hybrid multimedia high data rates; (ii) multiple simultaneous multimedia mmWave-Li-Fi connections; (iii) easier mmWave-Li-Fi ultra-low latency support; and (iv) operational adaptive beamforming modes for variable multimedia data rate and latency spanning short- and long-range 5G/6G vehicular communication use cases/applications [10], [20]. The future care prices will be influenced by the leading mmWave autonomous automotive use case and/or application domains including V2X for traffic efficiency; V2X for advanced driver assistance systems; V2X for fully automated driving [21], [20], [19].

3 Data Collection and Preprocessing

3.1 Data Collection

The dataset used in this study was scraped from Autotrader's website (Table 1). The dataset comprises 12 columns (features) and 402,005 rows, with certain features containing missing values and outliers that need to be addressed before constructing a machine-learning model.

Table 1: Analysis of Distribution

Feature	**Count**	**Unique**	**Top**	**Freq**	**Mean**	**Std**	**Min**	**25%**	**50%**	**75%**	**Max**
mileage	401878	NaN	NaN	NaN	37743.60	34831.72	0.00	10481.00	28629.50	56875.75	999999.00
reg code	370148	72	17	36738	NaN	NaN	NaN	NaN	NaN	NaN	NaN
colour	396627	22	Black	86287	NaN	NaN	NaN	NaN	NaN	NaN	NaN
make	402005	110	BMW	37376	NaN	NaN	NaN	NaN	NaN	NaN	NaN
model	402005	1168	Golf	11563	NaN	NaN	NaN	NaN	NaN	NaN	NaN
condition	402005	2	USED	370756	NaN	NaN	NaN	NaN	NaN	NaN	NaN
year of registra-tion	368694	NaN	NaN	NaN	2015.01	7.96	999.00	2013.00	2016.00	2018.00	2020.00
price	402005	NaN	NaN	NaN	17341.97	46437.46	120.00	7495.00	12600.00	20000.00	9999999.00
body type	401168	16	Hatch back	167315	NaN	NaN	NaN	NaN	NaN	NaN	NaN
car van	402005	2	False	400210	NaN	NaN	NaN	NaN	NaN	NaN	NaN
fuel type	401404	9	Petrol	216929	NaN	NaN	NaN	NaN	NaN	NaN	NaN

3.2 Data Preprocessing

The provided descriptive statistics in Table 1 offer insights into the central tendency, spread, and distribution of the dataset. Notably, the analysis suggests that the majority of vehicles are used rather than new. Additionally, black emerges as a popular colour choice, and petrol-powered vehicles appear to be a prevalent preference within the dataset. The distributions of mileage and price, as depicted below (Figure 1 and Figure 2), display positive skewness, as indicated by the mean surpassing the median. Remarkably, both variables exhibit the presence of outliers, particularly noticeable in the maximum values. The descriptive statistics for both price and mileage presented below reveal a high standard deviation, signifying a significant dispersion of values from the mean.

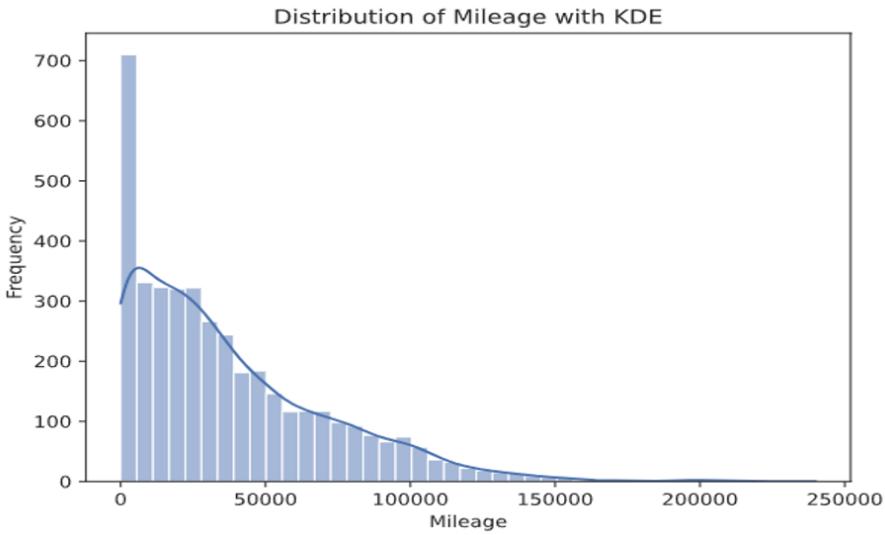

Fig. 1: Distribution of mileage

For the colour variable, the dataset below indicates that the most common car colours are Black, White, Grey, Blue, Silver, and Red. Conversely, less frequent occurrences of colours such as Burgundy, Magenta, Navy, and Indigo suggest that these hues are uncommon choices for car colours.

Regarding fuel types, the analysis reveals a predominance of traditional fuels such as Petrol and Diesel, constituting the majority of vehicle sales (54% with Petrol). Less conventional fuel types like Bi Fuel and Diesel Plug-in Hybrid have limited representation, while Natural Gas emerges as the rarest fuel type in the dataset.

4 Distribution of Mileage with Price

According to the correlation heatmap (Fig 5), 'mileage' exhibits a strong negative correlation with price. Hence, this confirms its significant predictive power for pricing. Fig 6 reveals a noticeable

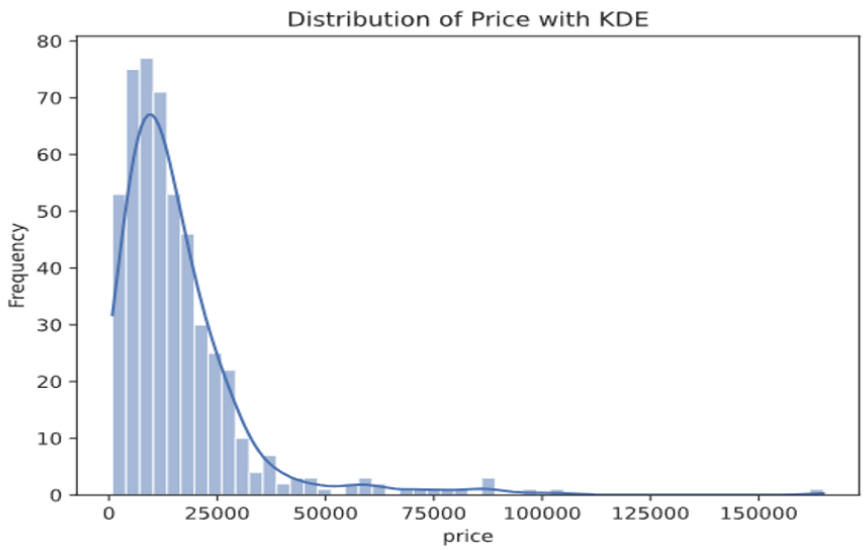

Fig. 2: Distribution of price

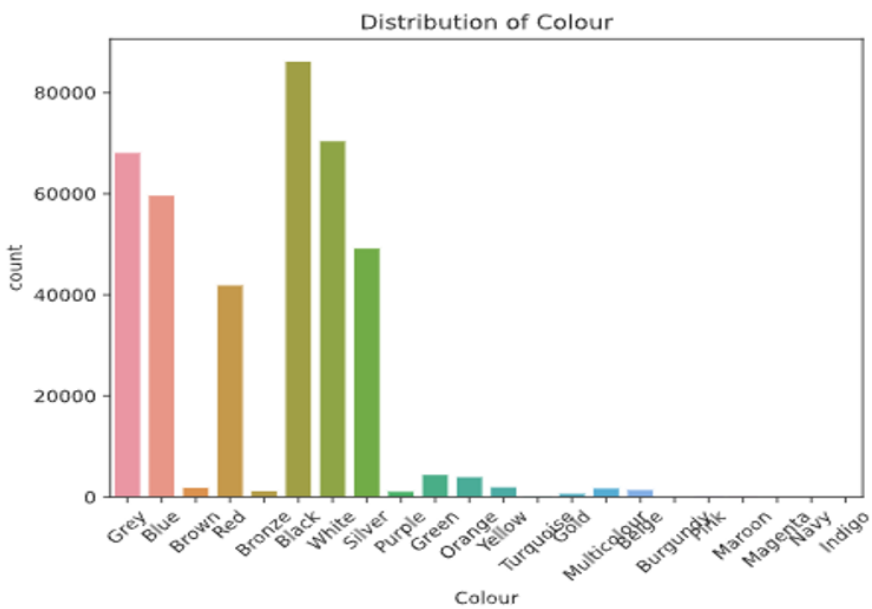

Fig. 3: Colour Choices

downward trend, indicating a potential correlation between 'mileage' and 'price.' This suggests that, as mileage increases, there is a general tendency for prices to decrease. This insight is critical

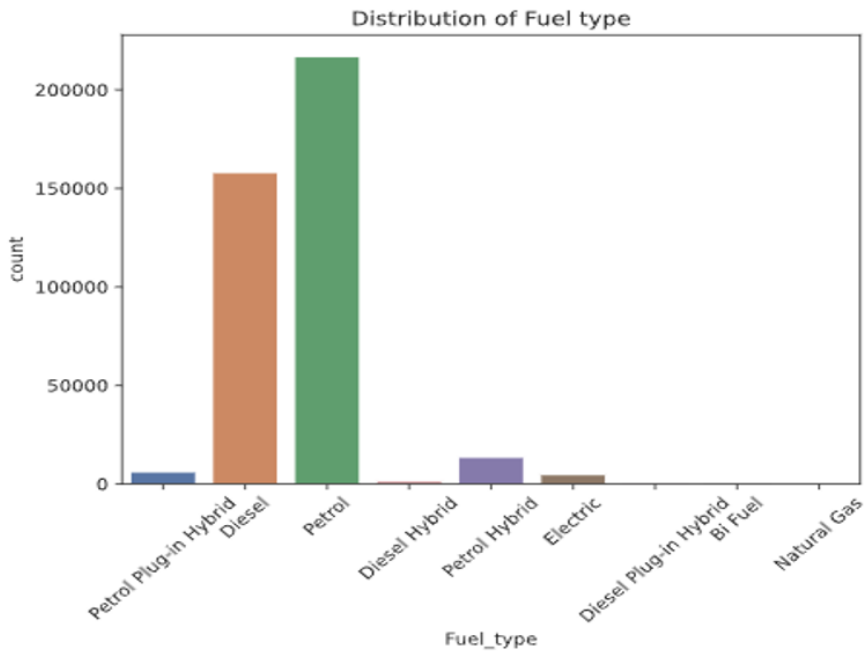

Fig. 4: Fuel choices

for evaluating the predictive capability of the 'mileage' feature in determining car prices. Fig 7 shows a bar plot comparing prices across the different colours. The plot shows that Black is the most common car colour and ranks as the fourth most expensive in a comparison of prices across different colours. This suggests that colour holds predictive power for determining car prices. But the correlation heatmap indicates a weak correlation between colour and price, emphasizing that colour alone is not a reliable metric for price prediction.

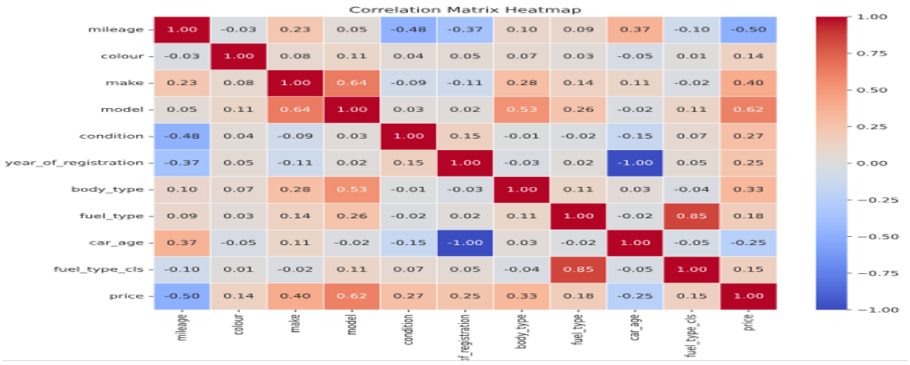

Fig. 5: Correlation Matrix and Heatmap for Visualizing Relationships Between Features and Target

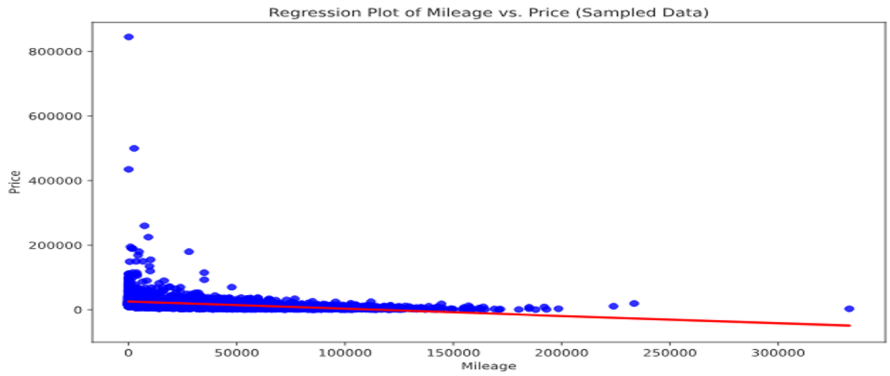

Fig. 6: Regression plot of Mileage against Price

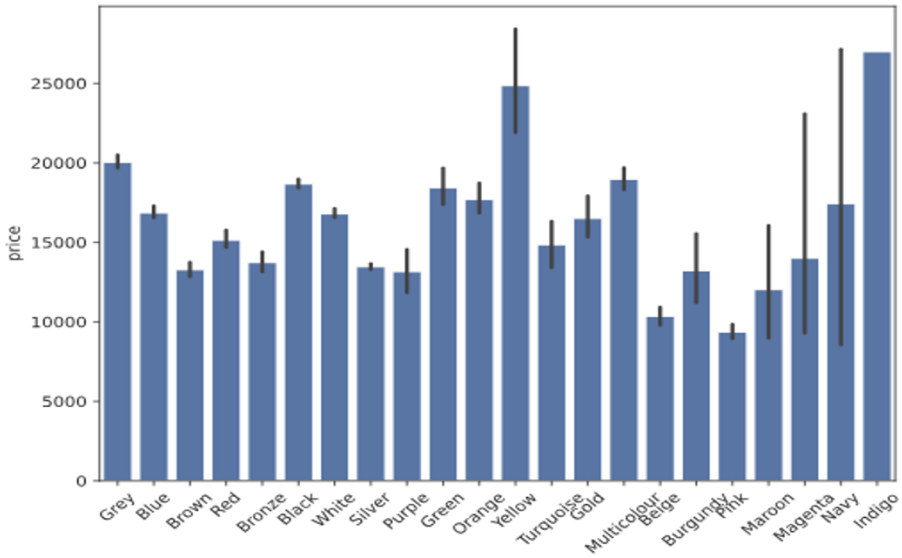

Fig. 7: Bar plot comparing prices across the different colours

4.1 Dealing With Incorrect Values in Mileage Column

Zero mileage for used cars is an inaccurate representation. As per information from Superkilometerfilter.com, this anomaly occurs when a used vehicle's odometer resets to zero after reaching a specific mileage threshold. For this analysis, such instances will be treated as inaccuracies, and accordingly, all zero values in the 'mileage' column for 'USED' vehicles will be substituted with the

computed median of the column. Additionally, the report examines vehicles with recorded mileage in the form of == 999999 to uncover potential inaccuracies. The investigation reveals that this value is an erroneous entry in the dataset for the specified car specification. Consequently, the decision was made to drop this entry.

4.2 Dealing With Incorrect Values in Year of Registration Column

As per CarReg, the initial series of UK plate numbers were introduced in 1903. Consequently, any recorded information predating 1903 is considered an incorrect value. To address this, it would be replaced with the mode of the year of registration column for values greater or equal to 1904. As part of the data cleaning process, the information from the UK car registration Wikipedia page has been used to fill in the missing values in the year of registration column using registration codes. According to information provided on the UK car registration Wikipedia page, the reg code is either the last two digits of the year itself, if issued between March and August (e.g. "12" for registrations issued between 1 March - 31 August 2012), or else has 50 added to that value if issued between September and February the following year(e.g "62" for registration issued between 1 September 2012 - 28/29 February 2013).

As part of the data cleaning process for machine learning, the decision was made to drop the reg code after using it to fill up for the year of registration, to avoid multicollinearity. Additionally, the 'public reference' column was excluded from consideration, as it functions as a unique identifier and is believed to contribute limited value to the analysis.

4.3 Evaluation Metrics

During the model evaluation and analysis, the performance of three chosen regression models—K Nearest Neighbors Regression (knnr), Decision Tree Regression (dtr), and Linear Regression (lr)—has been assessed. The models were evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2) (Fig 8). The Decision Tree Regression model exhibits a notable high-test score of 84%, indicating robust predictive performance. It is followed by the K Nearest Neighbors Regression model with 83%, and the Linear Regression model demonstrates the lowest score at 71%. These metrics provided a comprehensive understanding of the model's performance.

The Mean Absolute Error (MAE) on the test set serves as a crucial metric for evaluating model performance, where a lower MAE indicates that, on average, the model's predictions are closer to the actual values. In the analysis, the Decision Tree Regression model stands out with the most favourable performance, having the lowest MAE of 1886.12. This implies that the model generates predictions with relatively small errors. The K Nearest Neighbors Regression model closely follows with a slightly higher MAE of 1984.12, indicating commendable predictive capability. In contrast, the Linear Regression model records the highest MAE of 2724.28, suggesting larger prediction errors compared to the other models. These performance distinctions are visually represented in the accompanying bar plot.

The KNN regression analysis (Fig 9) of feature importance reveals that 'condition' holds the highest influence, followed by 'model,' 'make,' 'colour,' and 'mileage.' This underscores the significance of these specific features in predicting the target variable, indicating their substantial impact on the model's predictions. In linear regression (Fig 11), feature importance is ascertained by analyzing the coefficients assigned to each feature in the model, with larger coefficients indicating a

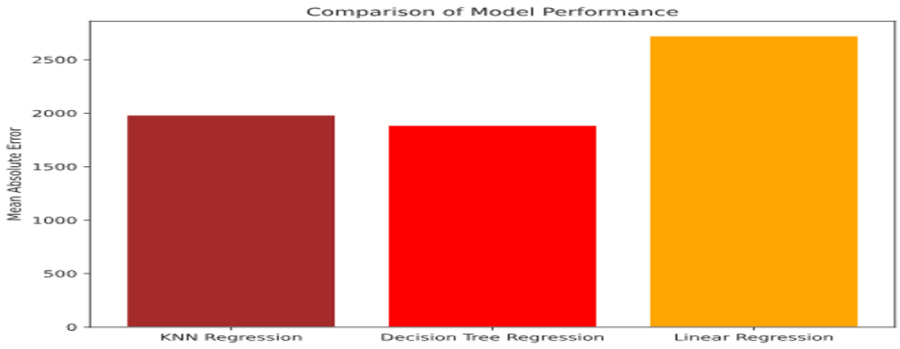

Fig. 8: Model performance comparison

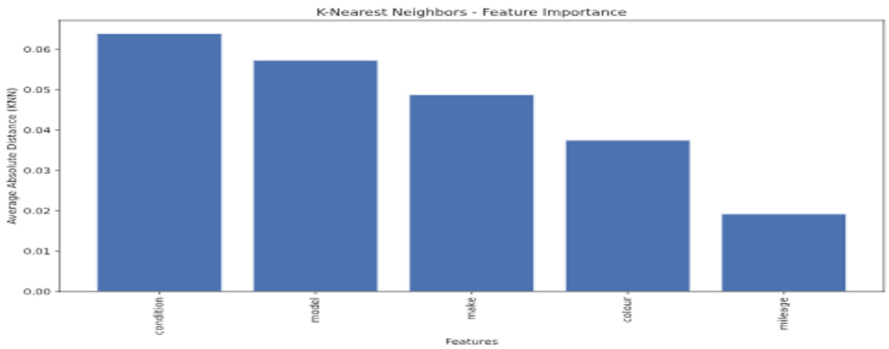

Fig. 9: KNN regression analysis

stronger impact on the target variable. The feature at the top of the plot is regarded as the most important feature in the linear regression model.

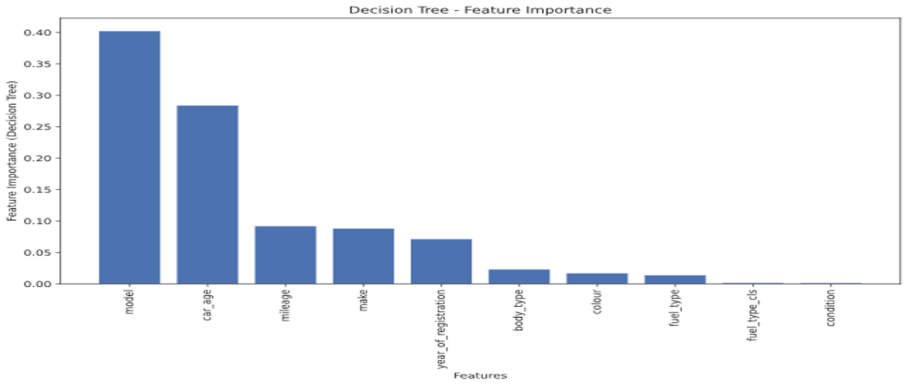

Fig. 10: Decision Tree Regression analysis

In the analysis of Decision Tree Regression (Fig 10), the evaluation of feature importance emphasizes the significance of 'model' as the primary factor, followed by 'car_age,' '$mileage$,' '$make$,' '$year_of_registration$,' '$body_type$,' and '$fuel_type$.' This highlights the influential role of these specific features in the decision-making process of the model.

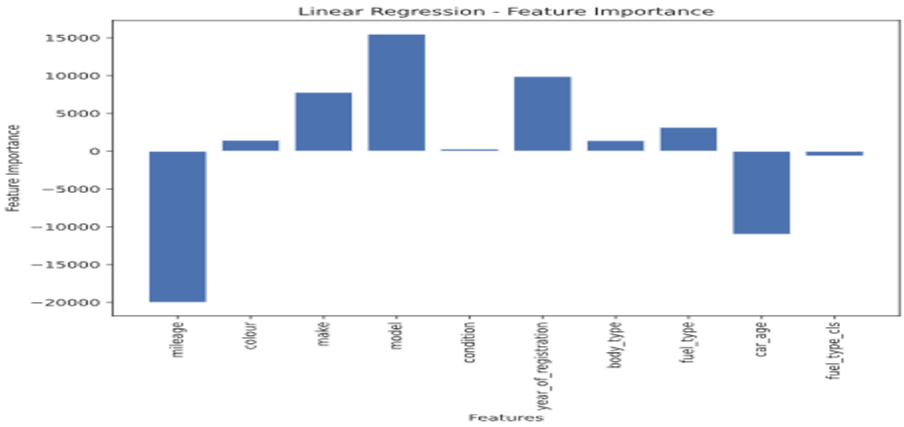

Fig. 11: Linear Regression analysis

5 Fine-Grained Evaluation

This entails a comprehensive evaluation of the performance of three algorithms on the car_df dataset. In True vs Predicted plots, where the x-coordinate represents the predicted value and the y-coordinate represents the actual value. The regplot incorporates a regression line on the plot, representing the best-fit regression line that illustrates the relationship between predicted and actual values. The fact that the line is at a 45-degree angle suggests a close alignment between predicted and actual values, indicating a well-performing model.

5.1 Fine-Grained Evaluation for KNeighbors Regressor

From Fig 12 data points clustered along the line suggest that the model is making accurate predictions for most portions of the data. The closeness to the line indicates that the predicted values are close to the actual values. This is a positive sign, suggesting that the model is capturing the underlying patterns in the data well. It indicates that the model is generally providing accurate predictions for instances that fall within the main cluster.

From Fig 13, the clustering of data points along the line for a Decision Tree Regressor implies accurate predictions, with their closeness indicating a strong alignment between predicted and actual values. Conversely, data points that diverge from the main cluster and are positioned at a distance from the line suggest challenges for the Decision Tree Regressor in accurately predicting those instances, possibly due to outliers and noise in the data. Instances marked by outliers or larger prediction errors requires model improvement.

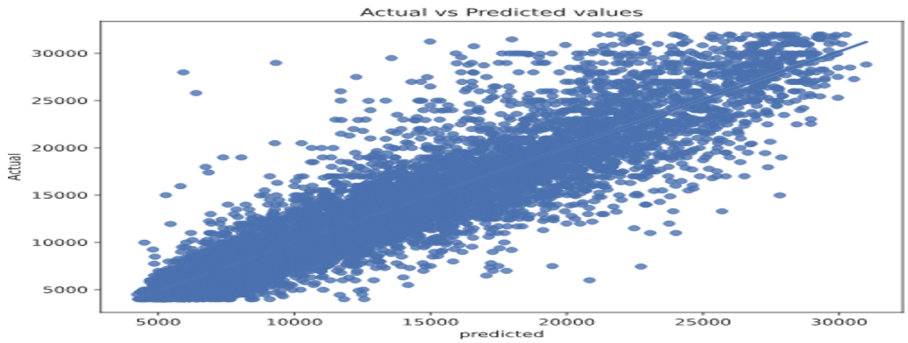

Fig. 12: Fine-Grained Evaluation for KNeighbors Regressor

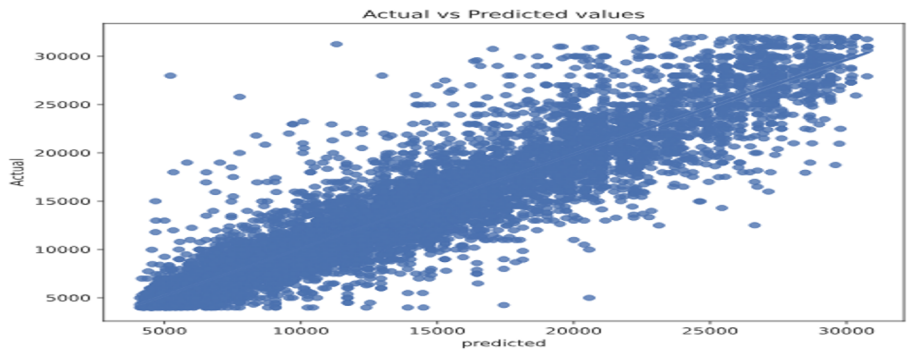

Fig. 13: Fine-Grained Evaluation for Decision Tree Regressor

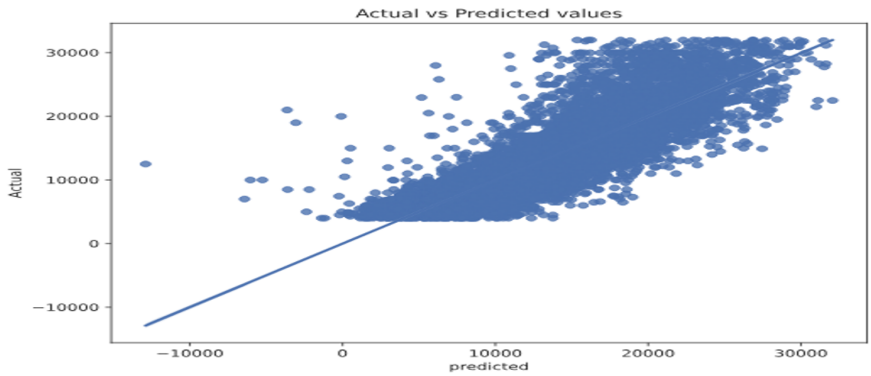

Fig. 14: Fine-Grained Evaluation for Linear Regression

From Fig 14, every point stands for a data point, and the Regplot includes a best-fit linear regression line showing the connection between predicted and actual values. The spread of points around this line is a visual representation of the variation in the model's predictions. a broader

spread indicates higher variability in its performance. Furthermore, outliers, those points situated far from the central cluster, may suggest instances where the model is not performing well. This requires further investigation.

The decision tree is probably the best-performing model because on average, most data points fall closer to the regression line.

6 Conclusion

The study successfully demonstrated the application of machine learning techniques to predict car prices on Autotrader. By incorporating a comprehensive set of features, including advanced functionalities such as multimedia infotainment and multi-sensor wireless connectivity capabilities, the predictive models achieved a significant improvement in accuracy over traditional methods. The Decision Tree Regression model emerged as the best predictor, offering valuable insights into the factors influencing car prices. The findings suggest that advanced machine learning algorithms can substantially enhance the reliability of car price estimations, benefiting consumers, dealers, and online car trading platforms. Future work could focus on incorporating additional features such as market trends and economic indicators to further improve the model's accuracy and robustness. Additionally, exploring the integration of AI-driven approaches and real-time data analytics could further enhance the predictive capabilities of these models, aligning with the evolving dynamics of the automotive market.

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